

Nassau County Sample — Shopping Center / Retail



Traffic Impact Study

Prepared August 18, 2026 · 40.8623, -73.6337

SIS sample draft — full tier

Traffic Impact Study

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TRAFFIC IMPACT STUDY FOR DOT PROJECTS

Prepared in Accordance With Chapter 5 of the NYSDOT Highway Design Manual (HDM)

Project name	Nassau County Sample — Shopping Center / Retail
PIN	PIN xxxx.xx (to be assigned by NYSDOT)
Project description	Shopping Center / Retail (85 1,000 sqft GLA) within New York-Newark-Jersey City MSA.
NYSDOT Region	Region 10 — Long Island
Site coordinates	40.8623°, -73.6337°
Opening year (ETC)	2027

Note:

It is a violation of law for any person, unless they are acting under the direction of a licensed professional engineer, architect, landscape architect, or land surveyor, to alter an item in any way. If an item bearing the stamp of a licensed professional is altered, the altering engineer, architect, landscape architect, or land surveyor shall stamp the document and include the notation "altered by" followed by their signature, the date of such alteration, and a specific description of the alteration.

This report is based on the NYSDOT TIS Shell revised on 9/16/2014.

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3.0 Project Specific Traffic Impacts Appendix A: Traffic Analysis Output

PROJECT CONTEXT

NYSDOT region: Region 10 — Long Island

Planning group: NYMTC (New York Metropolitan Transportation Council)

Regulatory regime: SEQRA (ECL Article 8 / 6 NYCRR Part 617).

SEQRA classification: Not yet determined — lead agency to classify as Type I (6 NYCRR Part 617.4 — likely EIS), Type II (Part 617.5 — exempt), or Unlisted (Part 617.6 — Short EAF + significance finding) before locking the TIS.

1.0 SUMMARY OF TRAFFIC IMPACTS

Capacity:

A capacity analysis was performed per NYSDOT Highway Design Manual Chapter 5. The following capacity improvement measures are cost effective, and are recommended to reduce reoccurring delay: Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback; Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback; Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback; Minor: signal-timing optimization (shift 3-5s of green to the critical phase) is sufficient. No geometric change required; Minor: signal-timing optimization (shift 3-5s of green to the critical phase) is sufficient. No geometric change required; Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback.

Crash:

A minimum three-year accident history review was performed and the safety screening met all of the required steps. A full crash analysis is not required.

GML §239-m / §239-n county planning board referral required:

The site is within 500 ft of GLEN COVE AVE (Urban Minor Arterial), which triggers mandatory referral to the controlling county planning board under General Municipal Law §239-m (site-plan / special-permit / variance / zoning amendment) or §239-n (subdivision review) before the local board may act. The county may make adverse findings under §239-f; the local board can override only by supermajority. Practical TIS consequence: the county will typically request the TIS in full as part of the §239 referral package — pre-application coordination with the county planning agency is

recommended.

20 INTERSECTIONS	3 LOS DROPS	4 AT LOS E/F	29.0s WORST Δ DELAY
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2.0 TRAVEL SPEEDS

Per NYSDOT HDM Chapter 5 §5.2, posted speed limits and actual operating speeds (85th-percentile) along the study network are reported below. Posted-speed-limit values in the third column are auto-ingested from the NYSDOT Roadway Data Management (RDM) FeatureServer (https://gis.dot.ny.gov/hostingny/rest/services/Roadways/RDM_Roadway_Current/FeatureServer) at PDF-generation time — the same source data viewable through the NYSDOT Traffic Data Viewer (<https://gis.dot.ny.gov/html5viewer/?viewer=tdv>). Operating speeds (85th-percentile) are NOT published by NYSDOT and require a project-specific radar / floating-car speed study.

Street	Functional class	Posted speed	85th-pctile operating speed
Glen Cove Avenue (GLEN COVE AVE)	Urban Minor Arterial	30 mph	Speed study required
Glen Cove Avenue (PRATT BLVD)	Urban Principal Arterial Other	30 mph	Speed study required
Brewster Street (GLEN COVE AVE)	Urban Minor Arterial	30 mph	Speed study required
Glen Cove Avenue (GLEN COVE AVE)	Urban Minor Arterial	30 mph	Speed study required
Near Charles Street (CHARLES ST)	Urban Major Collector	30 mph	Speed study required
Bridge Street (BRIDGE SCHL ST)	Urban Minor Arterial	Not posted (local)	Speed study required

3.0 CAPACITY ANALYSIS

Capacity analyses performed in this report are consistent with HCM 6th Edition. The software used to perform this analysis is the SimpleImpactStudies HCM 6 solver (per-approach control-delay model). Synchro / SimTraffic, HCS, Sidra, or VISSIM may be substituted for formal submittal per NYSDOT Regional Traffic Office preference.

Study network functional-class composition (per NYSDOT RDM): Urban Minor Arterial (11); Urban Major Collector (5); Urban Principal Arterial Other (3); Urban Local (1). NYSDOT functional class drives access management expectations under HDM Appendix 5A (entrance standards for state highways) and the default K-factor banding under HDM §5.3.

Level of Service thresholds applied below are per HCM 2010 Exhibits 18-4 (signalized), 19-1 / 20-2 / 21-1 (unsignalized — 2-way stop / all-way stop / roundabout), and 10-7 (freeway). Any component $v/c > 1.0$ is LOS F regardless of delay or density.

Signalized intersections (HCM 2010 Exhibit 18-4, p. 18-6) — control delay sec/veh; any $v/c > 1.0$ is F.

LOS	A	B	C	D	E	F
sec/veh	≤10	10-20	20-35	35-55	55-80	>80

Unsignalized — 2-way stop / all-way stop / roundabout (HCM 2010 Exhibits 19-1, 20-2, 21-1) — sec/veh; any $v/c > 1.0$ is F.

LOS	A	B	C	D	E	F
sec/veh	≤10	10-15	15-25	25-35	35-50	>50

Freeway basic / weaving / merge-diverge (HCM 2010 Exhibit 10-7, p. 10-9) — density pc/mi/ln; any component $vd/c > 1.00$ is F.

LOS	A	B	C	D	E	F
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LOS	A	B	C	D	E	F
pc/mi/ln	≤11	11-18	18-26	26-35	35-45	>45

3.1 Growth Rates

Background traffic growth is applied at 1.5% per year over 1 year. This rate is the per-station median CAGR measured from the NYSDOT Traffic Monitoring AADT FeatureServer for Region 10 — Long Island: 0.07%/yr median across 12,244 count stations (rolling actual-count window approximately 2000-2019; 25th-75th percentile range -1.57% to 1.80%). The measured value supersedes the NYSDOT statewide default fallbacks (per HDM Chapter 5: 1.7%/yr Interstate + ramps from the Data Services Bureau, 2.0%/yr all other roads from the Regional Planning Group). The controlling Regional Planning Group NYMTC (New York Metropolitan Transportation Council) may have separately-adopted growth-rate guidance that overrides the measured median for formal submittal — confirm at scoping.

3.2 Existing AADT and DHV

Existing AADT is sourced from the NYSDOT Traffic Data Viewer. Design Hourly Volume (DHV) is reported per HDM Chapter 5 convention as AADT × K-factor; this analysis applies K = 0.10 as the statewide design default — facility-specific K should be substituted from NYSDOT count-station records for formal submittal. The table below grows the existing volume at 1.5%/yr to Estimated Time of Completion (ETC), ETC + 20 years, and ETC + 30 years per HDM Chapter 5 §5.3.

Intersection	2026 AADT	2026 DHV	2027 (ETC) AADT	2027 (ETC) DHV	2047 AADT	2047 DHV	2057 AADT	2057 DHV
Glen Cove Avenue & Charles Street	17,130	1,713	17,387	1,739	23,418	2,342	27,177	2,718
Glen Cove Avenue & Bridge Street	3,049	305	3,095	309	4,168	417	4,837	484
Brewster Street & Herb Hill Road	17,130	1,713	17,387	1,739	23,418	2,342	27,177	2,718
Glen Cove Avenue & Charles Street	17,130	1,713	17,387	1,739	23,418	2,342	27,177	2,718
Near Charles Street	1,170	117	1,188	119	1,599	160	1,856	186
Bridge Street & School Street	3,061	306	3,107	311	4,185	418	4,856	486
Pratt Boulevard & Glen Street	10,199	1,020	10,352	1,035	13,943	1,394	16,181	1,618
Brewster Street & School Street	10,451	1,045	10,608	1,061	14,287	1,429	16,581	1,658
School Street & Highland Road	1,001	100	1,016	102	1,368	137	1,588	159
Shore Road & Glen Cove Avenue	17,131	1,713	17,388	1,739	23,419	2,342	27,179	2,718
Glen Street & Pratt Boulevard	3,060	306	3,106	311	4,183	418	4,855	485
Pratt Boulevard & Town Path	11,750	1,175	11,926	1,193	16,063	1,606	18,642	1,864
School Street & Brewster Street	3,670	367	3,725	373	5,017	502	5,823	582
Landing Road & Cottage Row	3,580	358	3,634	363	4,894	489	5,680	568
Glen Street & Town Path	3,060	306	3,106	311	4,183	418	4,855	485
Near Glen Cove Avenue	17,130	1,713	17,387	1,739	23,418	2,342	27,177	2,718
Dosoris Lane & Forest Avenue	1,150	115	1,167	117	1,572	157	1,825	182
Glen Street & Pearsall Avenue	1,649	165	1,674	167	2,254	225	2,616	262
Near Forest Avenue	14,960	1,496	15,184	1,518	20,451	2,045	23,734	2,373
Cedar Swamp Road & Glen Street	18,610	1,861	18,889	1,889	25,441	2,544	29,525	2,953

Source: AADT estimated from engine peak-hour entering-volume sum via K = 0.10. NYSDOT Traffic Data Viewer measured AADT is recommended for submittal. Projection applies the 1.5%/yr background growth rate compounded annually.

3.3 Traffic Control Device Data

Existing traffic control devices at each study-network intersection are summarized below. Signal timing data (cycle length, phase splits, offset) and per-approach detection presence should be confirmed against the controlling NYSDOT Regional Traffic Office or municipal traffic-engineering signal record for formal submittal.

Intersection	Control type	Approaches w/ detection
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Intersection	Control type	Approaches w/ detection
Glen Cove Avenue & Charles Street	Signal (controlling RTO to confirm)	Field verification required
Glen Cove Avenue & Bridge Street	Signal (controlling RTO to confirm)	Field verification required
Brewster Street & Herb Hill Road	Signal (controlling RTO to confirm)	Field verification required
Glen Cove Avenue & Charles Street	Signal (controlling RTO to confirm)	Field verification required
Near Charles Street	Signal (controlling RTO to confirm)	Field verification required
Bridge Street & School Street	Signal (controlling RTO to confirm)	Field verification required
Pratt Boulevard & Glen Street	Signal (controlling RTO to confirm)	Field verification required
Brewster Street & School Street	Signal (controlling RTO to confirm)	Field verification required
School Street & Highland Road	Signal (controlling RTO to confirm)	Field verification required
Shore Road & Glen Cove Avenue	Signal (controlling RTO to confirm)	Field verification required
Glen Street & Pratt Boulevard	Signal (controlling RTO to confirm)	Field verification required
Pratt Boulevard & Town Path	Signal (controlling RTO to confirm)	Field verification required
School Street & Brewster Street	Signal (controlling RTO to confirm)	Field verification required
Landing Road & Cottage Row	Signal (controlling RTO to confirm)	Field verification required
Glen Street & Town Path	Signal (controlling RTO to confirm)	Field verification required
Near Glen Cove Avenue	Signal (controlling RTO to confirm)	Field verification required
Dosoris Lane & Forest Avenue	Signal (controlling RTO to confirm)	Field verification required
Glen Street & Pearsall Avenue	Signal (controlling RTO to confirm)	Field verification required
Near Forest Avenue	Signal (controlling RTO to confirm)	Field verification required
Cedar Swamp Road & Glen Street	Signal (controlling RTO to confirm)	Field verification required

3.4 Capacity Analysis for Existing No-Build Condition

Existing No-Build conditions reflect current-year peak-hour volumes (no growth, no project trips). Existing measured volumes are grown to opening year 2027 at 1.5%/yr to establish the No-Build baseline for §3.5 comparison.

Intersection	Existing LOS	Existing delay (s)	No-Build LOS	No-Build delay (s)	v/c
Glen Cove Avenue & Charles Street	D	45.6	D	48.4	0.97
Glen Cove Avenue & Bridge Street	B	15.2	B	15.2	0.17
Brewster Street & Herb Hill Road	D	45.6	D	48.4	0.97
Glen Cove Avenue & Charles Street	D	45.6	D	48.4	0.97
Near Charles Street	B	14.2	B	14.2	0.07
Bridge Street & School Street	B	15.2	B	15.2	0.17
Pratt Boulevard & Glen Street	C	21.1	C	21.3	0.58
Brewster Street & School Street	C	21.5	C	21.7	0.59
School Street & Highland Road	B	14.1	B	14.1	0.06
Shore Road & Glen Cove Avenue	D	45.6	D	48.4	0.97
Glen Street & Pratt Boulevard	B	15.2	B	15.2	0.17
Pratt Boulevard & Town Path	C	23.3	C	23.6	0.66
School Street & Brewster Street	B	15.6	B	15.6	0.21
Landing Road & Cottage Row	B	15.5	B	15.5	0.20

Intersection	Existing LOS	Existing delay (s)	No-Build LOS	No-Build delay (s)	v/c
Glen Street & Town Path	B	15.2	B	15.2	0.17
Near Glen Cove Avenue	D	45.6	D	48.4	0.97
Dosoris Lane & Forest Avenue	B	14.2	B	14.2	0.06
Glen Street & Pearsall Avenue	B	14.4	B	14.4	0.09
Near Forest Avenue	C	31.4	C	32.4	0.84
Cedar Swamp Road & Glen Street	E	65.2	E	69.9	1.05

Trip Distribution by Time of Day

Estimated within-day distribution of the 6,800 gross daily trips and the resulting on-site accumulation. Within-day distribution from NHTS 2022 shopping / errand / meal trip start times, cross-checked against public-data time-of-day patterns for LU 820 / 850. Activity builds to a midday plateau and a stronger late-afternoon/evening peak; on-site occupancy peaks midday through the evening.

Figure — Inbound / Outbound Trips by Start Time

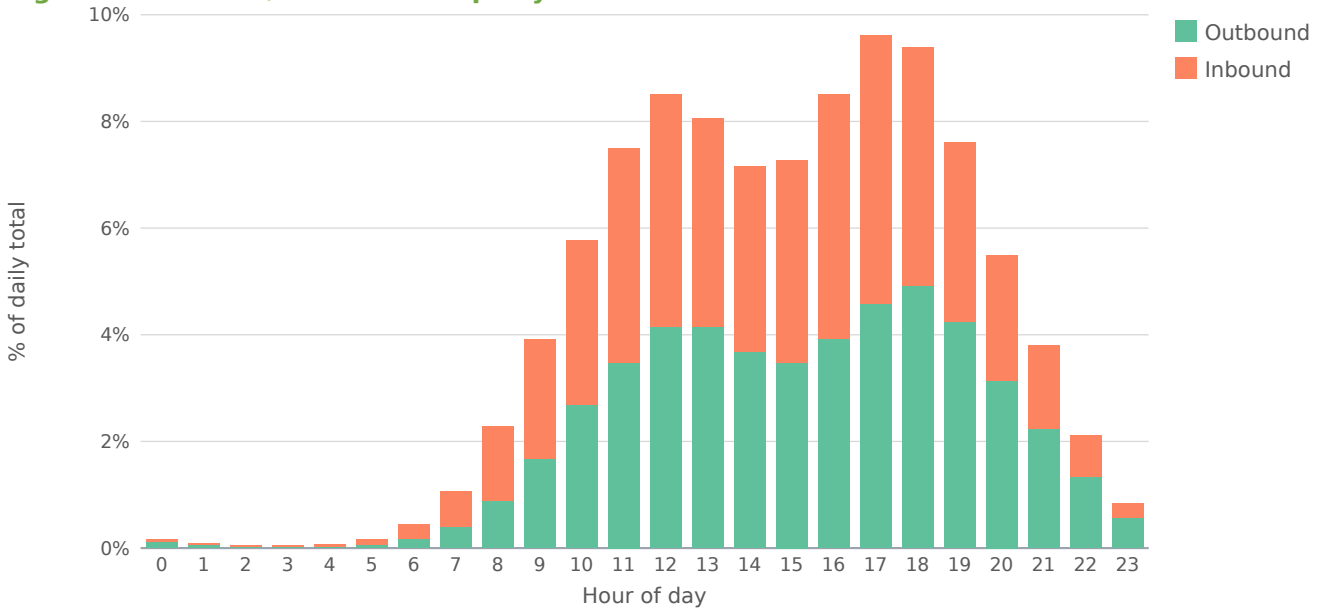
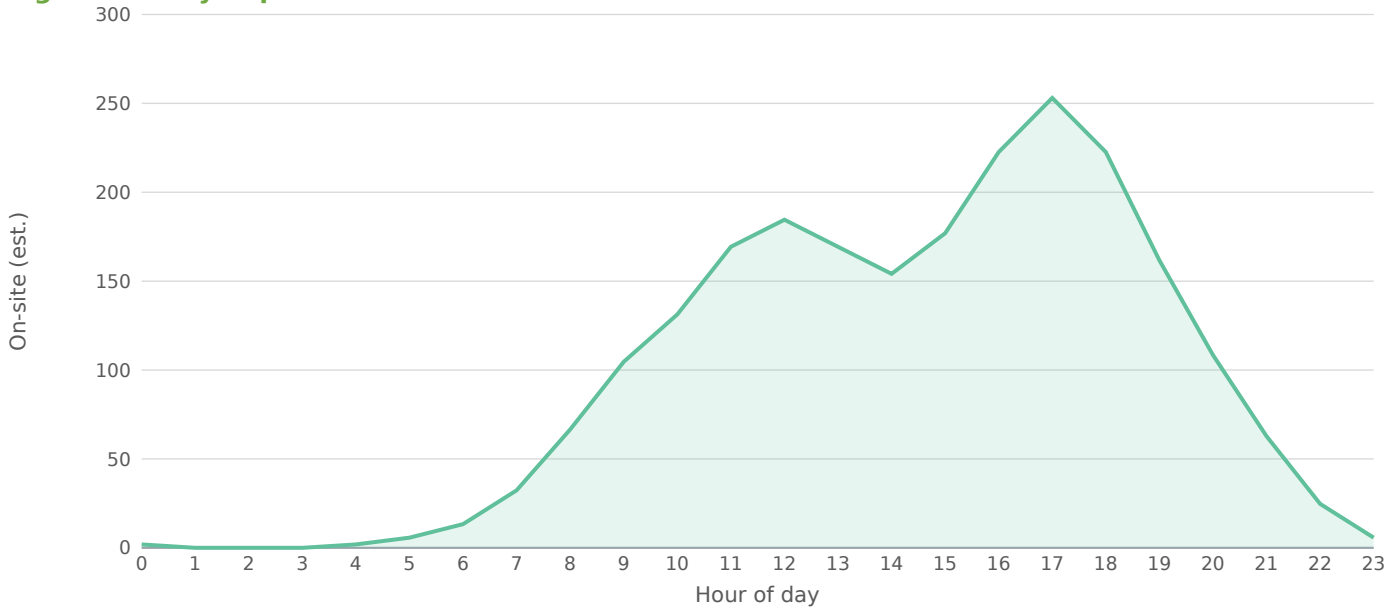


Figure — Daily Trip Accumulation



Peak on-site accumulation ~253 at 17:00, from 6,800 gross daily trips on the retail within-day profile. Screening estimate; not a substitute for a calibrated time-of-day model.

3.5 Capacity Analysis for Proposed Build Condition

Build conditions add the project's external peak-hour trips (680 PM peak, 326 inbound / 354 outbound) to the No-Build baseline at each affected intersection. Per HDM Chapter 5, intersection projects should also analyze each adjacent intersection plus any signal within ½ mile of the project site if expected left-turn volume changes materially affect capacity. The engine's network is determined by a 0.75-mile screening radius; manual scoping should confirm the ½-mile left-turn rule is satisfied.

Intersection	ETC No-Build	ETC Build	ETC+20 NB	ETC+20 Bld	Δ delay (s)
Glen Cove Avenue & Charles Street	D	▲ E	F	F	29.0
Glen Cove Avenue & Bridge Street	B	B	B	B	0.8
Brewster Street & Herb Hill Road	D	▲ E	F	F	17.4
Glen Cove Avenue & Charles Street	D	▲ E	F	F	13.9
Near Charles Street	B	B	B	B	0.5
Bridge Street & School Street	B	B	B	B	0.6
Pratt Boulevard & Glen Street	C	C	C	C	1.0
Brewster Street & School Street	C	C	C	C	0.6
School Street & Highland Road	B	B	B	B	0.2
Shore Road & Glen Cove Avenue	D	D	F	F	4.1
Glen Street & Pratt Boulevard	B	B	B	B	0.2
Pratt Boulevard & Town Path	C	C	D	D	0.5
School Street & Brewster Street	B	B	B	B	0.1
Landing Road & Cottage Row	B	B	B	B	0.2
Glen Street & Town Path	B	B	B	B	0.1
Near Glen Cove Avenue	D	D	F	F	2.0

Intersection	ETC No-Build	ETC Build	ETC+20 NB	ETC+20 Bld	Δ delay (s)
Dosoris Lane & Forest Avenue	B	B	B	B	0.0
Glen Street & Pearsall Avenue	B	B	B	B	0.1
Near Forest Avenue	C	C	F	F	0.3
Cedar Swamp Road & Glen Street	E	E	F	F	1.0

3.6 Capacity Improvement Measures

The following capacity improvements are recommended at affected intersections under Build conditions. Improvement category guidance per HDM Chapter 5: addition of turn lanes; installation or re-timing of traffic signals; signage and striping to prevent unwanted turning movements; conversion to roundabout; and TDM / TSM measures per HDM Chapter 24.

Glen Cove Avenue & Charles Street [MODERATE]

Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback.

Brewster Street & Herb Hill Road [MODERATE]

Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback.

Glen Cove Avenue & Charles Street [MODERATE]

Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback.

Shore Road & Glen Cove Avenue [MINOR]

Minor: signal-timing optimization (shift 3-5s of green to the critical phase) is sufficient. No geometric change required.

Near Glen Cove Avenue [MINOR]

Minor: signal-timing optimization (shift 3-5s of green to the critical phase) is sufficient. No geometric change required.

Cedar Swamp Road & Glen Street [MODERATE]

Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback.

3.7 Trip Distribution — Gravity Model

The trip distribution uses the gravity model (term = $M / (d^{1.1} \cdot d_{site})$; mass basis: intersection through-volume (AADT × K-factor) as the destination-activity attraction proxy). Shares are normalized to 100% of project trips and drive the trip assignment below. Basis: NCHRP-716 gamma-friction gravity model (production-constrained) over study-area attraction zones.

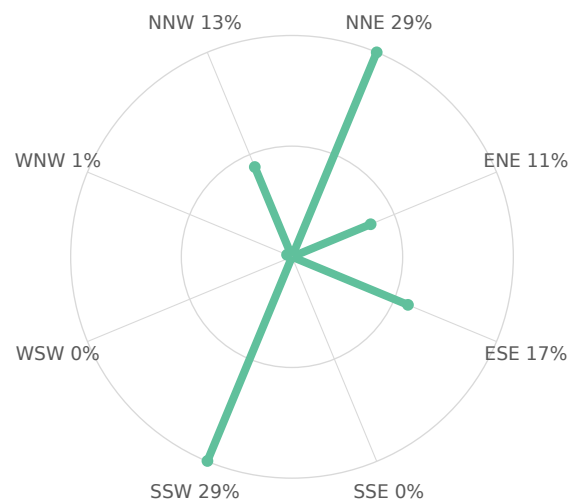
Directional sector	Share of project trips
Northeast (NNE + ENE)	41%
Southeast (ESE + SSE)	17%
Southwest (SSW + WSW)	29%
Northwest (WNW + NNW)	14%

Study-area zone	Dir.	Distance (mi)	Mass (M)	Gravity term	Trip share
Glen Cove Avenue & Charles Street	NNW	0.02	1,713	0.1	10.91%
Brewster Street & Herb Hill Road	NNE	0.10	1,713	0.1	10.66%

Study-area zone	Dir.	Distance (mi)	Mass (M)	Gravity term	Trip share
Glen Cove Avenue & Charles Street	SSW	0.14	1,713	0.1	10.47%
Shore Road & Glen Cove Avenue	SSW	0.37	1,713	0.1	9.60%
Cedar Swamp Road & Glen Street	ESE	0.72	1,861	0.1	9.27%
Near Glen Cove Avenue	SSW	0.50	1,713	0.1	9.17%
Near Forest Avenue	NNE	0.67	1,496	0.1	7.59%
Pratt Boulevard & Town Path	ESE	0.42	1,175	0.1	6.48%
Pratt Boulevard & Glen Street	ENE	0.22	1,020	0.1	6.04%
Brewster Street & School Street	NNE	0.30	1,045	0.1	6.00%
School Street & Brewster Street	NNE	0.43	367	0.0	2.02%
Landing Road & Cottage Row	NNW	0.44	358	0.0	1.96%
Glen Cove Avenue & Bridge Street	ENE	0.10	305	0.0	1.90%
Bridge Street & School Street	NNE	0.16	306	0.0	1.86%
Glen Street & Pratt Boulevard	ENE	0.41	306	0.0	1.69%
Glen Street & Town Path	ENE	0.50	306	0.0	1.64%
Glen Street & Pearsall Avenue	ESE	0.58	165	0.0	0.86%
Near Charles Street	WNW	0.16	117	0.0	0.71%
Dosoris Lane & Forest Avenue	NNE	0.54	115	0.0	0.61%
School Street & Highland Road	NNE	0.31	100	0.0	0.57%

20 study-area zone(s). Screening-grade distribution; not a calibrated regional model.

Figure — Directional Distribution of Project Trips



Screening-grade directional distribution of net new project trips by compass octant (spoke length $\propto$ percent of project trips). Derived from the Gravity Model distribution.

Figure — Project Trip Share by Study-Area Zone

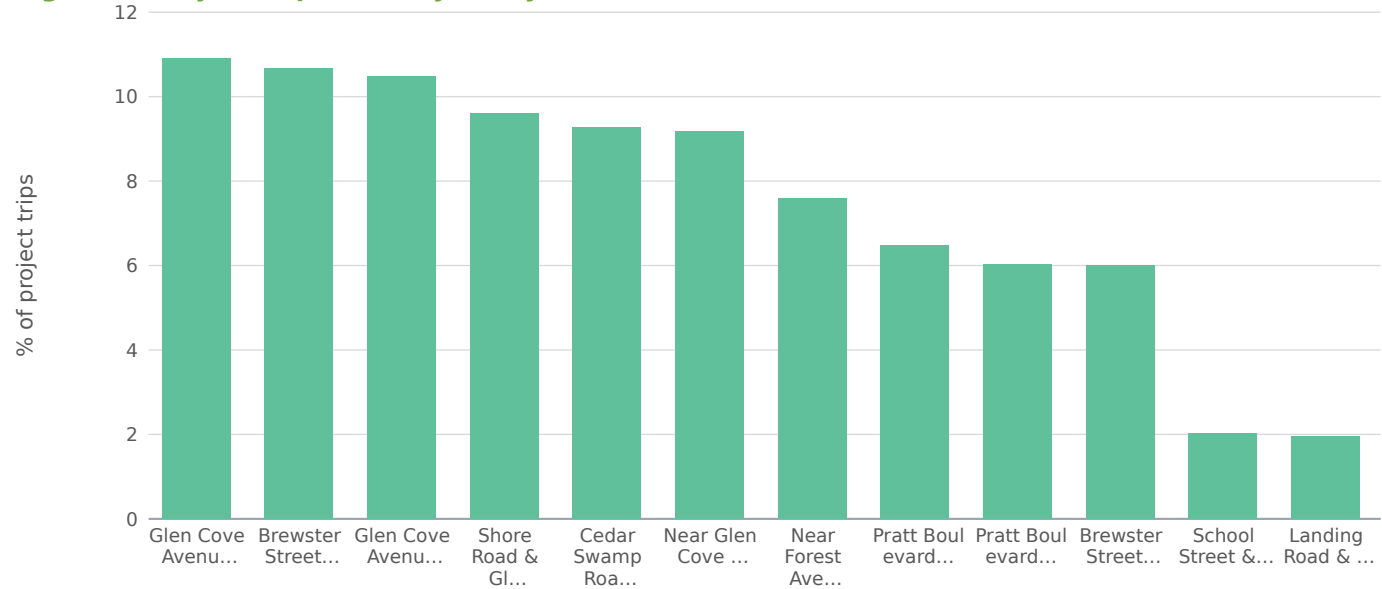
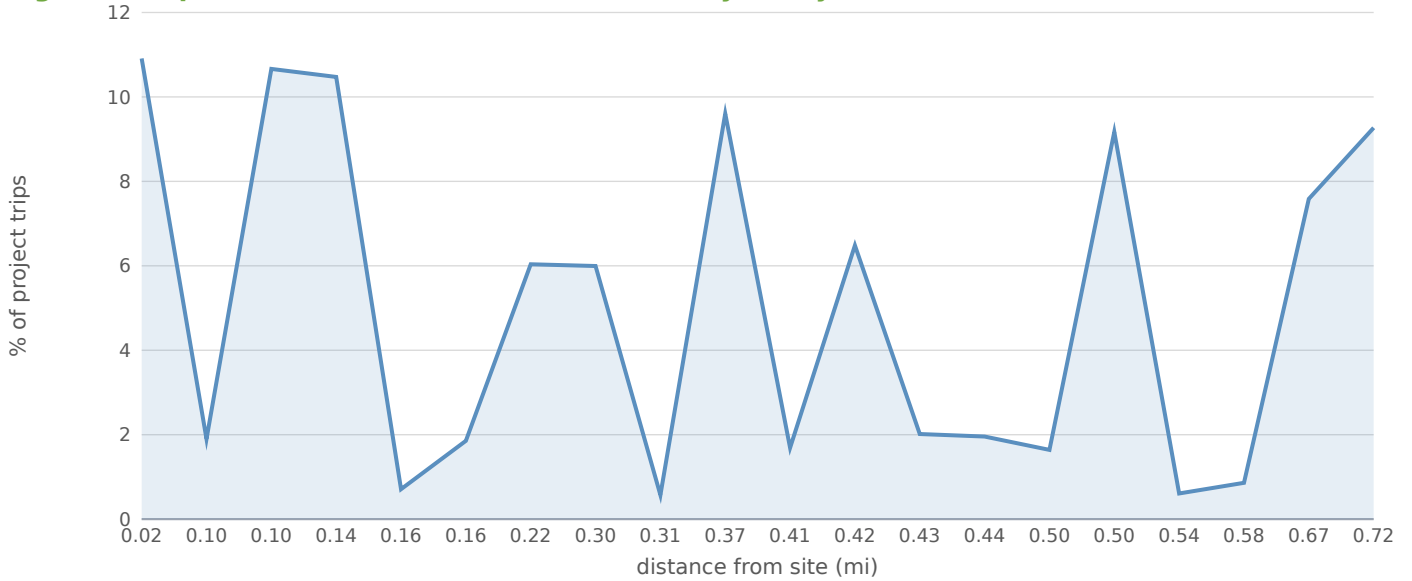


Figure — Trip Share vs. Distance from Site (Gravity Decay)



3.8 Project Trip Assignment

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
Glen Cove Avenue & Charles Street	NNW	0.02	101	191	17.4%
Glen Cove Avenue & Bridge Street	ENE	0.10	67	126	11.5%
Brewster Street & Herb Hill Road	NNE	0.10	67	126	11.5%
Glen Cove Avenue & Charles Street	SSW	0.14	55	104	9.5%
Near Charles Street	WNW	0.16	50	94	8.5%
Bridge Street & School Street	NNE	0.16	50	94	8.5%
Pratt Boulevard & Glen Street	ENE	0.22	37	69	6.3%
Brewster Street & School Street	NNE	0.30	24	46	4.2%
School Street & Highland Road	NNE	0.31	23	44	4.0%
Shore Road & Glen Cove Avenue	SSW	0.37	18	33	3.0%
Glen Street & Pratt Boulevard	ENE	0.41	14	27	2.5%
Pratt Boulevard & Town Path	ESE	0.42	14	26	2.4%
School Street & Brewster Street	NNE	0.43	13	25	2.3%
Landing Road & Cottage Row	NNW	0.44	12	23	2.1%
Glen Street & Town Path	ENE	0.50	9	17	1.5%
Near Glen Cove Avenue	SSW	0.50	9	17	1.5%
Dosoris Lane & Forest Avenue	NNE	0.54	7	14	1.3%

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
Glen Street & Pearsall Avenue	ESE	0.58	6	11	1.0%
Near Forest Avenue	NNE	0.67	4	7	0.6%
Cedar Swamp Road & Glen Street	ESE	0.72	3	6	0.5%

Project trips assigned to each study intersection from the distribution shares above: the directional shares orient the loading toward the high-share sectors, distance-decayed to each approach.

4.0 CRASH ANALYSIS

A minimum three-year accident history review was performed for the period 2023 to 2026 per HDM Chapter 5 §5.3.4. County-level crash counts below are auto-ingested from the New York State Police "Motor Vehicle Crashes – Case Information: Three Year Window" Open Data dataset (<https://data.ny.gov/Transportation/Motor-Vehicle-Crashes-Case-Information-Three-Year-/e8ky-4vqe>), filtered by county_name resolved from the project site coordinates via the NYSDOT RDM_Roadway_Current FeatureServer. The values cover the entire rolling 3-year reporting window for NASSAU County, statewide-system-wide — they are NOT a per-segment or per-intersection rate.

Severity descriptor (NY DMV)	Count (3-yr rolling)	Annualized
Fatal Accident	280	93.3
Injury Accident	6,376	2125.3
Property Damage Accident	113,811	37937.0
Property Damage & Injury Accident	35,025	11675.0
Total (all severities)	155,492	51830.7

Within NASSAU County over the rolling 3-year window, 155,492 crashes were reported to NY DMV (280 fatal, 41,401 injury-involved, 113,811 property damage only). The Fatal share is 0.060% of all reported crashes (annualized). For the Safety Screening per HDM Chapter 5 §5.3.4, the project-specific rate must be normalized against the controlling facility's exposure (segment AADT × segment length; intersection AADT × entry-flow) for comparison to NYSDOT Office of Modal Safety statewide averages and the 1.5× HAL trigger. County-totals above provide a county-baseline reference only — site-radius (1/4-mile typical) crash records, HAL / PIL / SDL flags, and CMF / CRF mitigation values must be obtained from the Region 10 — Long Island Regional Traffic Office via FOIL or SIMS. Required NYSDOT forms when the §4.0 section is expanded for formal submittal: TE-156a (Collision Diagram), TE-164a (Safety Benefits Evaluation), TE-204a (Accident Rate Summary), TE-213 (HAL / PIL / SDL Screening Output).

APPENDIX A — EXISTING VOLUME REPORT

Per-approach existing peak-hour volumes (vph) for all affected intersections within study limits.
Source: SimpleImpactStudies HCM 6 solver, calibrated against the controlling DOT 511 / Regional Traffic Office data feed.

Intersection	Approach	Existing (vph)	No-Build (vph)	Build (vph)
Glen Cove Avenue & Charles Street	NB	491	499	598
Glen Cove Avenue & Charles Street	SB	479	486	577
Glen Cove Avenue & Charles Street	EB	367	373	375
Glen Cove Avenue & Charles Street	WB	375	381	381
Glen Cove Avenue & Bridge Street	NB	89	90	90
Glen Cove Avenue & Bridge Street	SB	78	79	109
Glen Cove Avenue & Bridge Street	EB	81	82	148
Glen Cove Avenue & Bridge Street	WB	57	58	89
Brewster Street & Herb Hill Road	NB	486	493	559
Brewster Street & Herb Hill Road	SB	462	469	522
Brewster Street & Herb Hill Road	EB	417	423	424
Brewster Street & Herb Hill Road	WB	349	354	361
Glen Cove Avenue & Charles Street	NB	529	537	582
Glen Cove Avenue & Charles Street	SB	441	448	502
Glen Cove Avenue & Charles Street	EB	413	419	419
Glen Cove Avenue & Charles Street	WB	329	334	340
Near Charles Street	NB	39	39	54
Near Charles Street	SB	28	29	57
Near Charles Street	EB	27	27	30
Near Charles Street	WB	23	23	72
Bridge Street & School Street	NB	80	82	130
Bridge Street & School Street	SB	79	80	113
Bridge Street & School Street	EB	81	82	82
Bridge Street & School Street	WB	66	67	80
Pratt Boulevard & Glen Street	NB	304	308	308
Pratt Boulevard & Glen Street	SB	228	232	245
Pratt Boulevard & Glen Street	EB	257	261	297
Pratt Boulevard & Glen Street	WB	231	235	254
Brewster Street & School Street	NB	336	341	365
Brewster Street & School Street	SB	279	284	302
Brewster Street & School Street	EB	228	232	232
Brewster Street & School Street	WB	202	205	208
School Street & Highland Road	NB	29	29	52
School Street & Highland Road	SB	24	25	41
School Street & Highland Road	EB	24	24	24

Intersection	Approach	Existing (vph)	No-Build (vph)	Build (vph)
School Street & Highland Road	WB	24	24	28
Shore Road & Glen Cove Avenue	NB	555	563	577
Shore Road & Glen Cove Avenue	SB	390	395	413
Shore Road & Glen Cove Avenue	EB	389	395	395
Shore Road & Glen Cove Avenue	WB	380	386	387
Glen Street & Pratt Boulevard	NB	92	93	93
Glen Street & Pratt Boulevard	SB	75	76	81
Glen Street & Pratt Boulevard	EB	73	74	88
Glen Street & Pratt Boulevard	WB	66	67	75
Pratt Boulevard & Town Path	NB	339	344	344
Pratt Boulevard & Town Path	SB	291	295	299
Pratt Boulevard & Town Path	EB	323	328	341
Pratt Boulevard & Town Path	WB	222	226	235
School Street & Brewster Street	NB	114	116	128
School Street & Brewster Street	SB	86	87	97
School Street & Brewster Street	EB	91	92	92
School Street & Brewster Street	WB	77	78	80
Landing Road & Cottage Row	NB	100	102	114
Landing Road & Cottage Row	SB	100	101	111
Landing Road & Cottage Row	EB	87	88	88
Landing Road & Cottage Row	WB	71	72	73
Glen Street & Town Path	NB	83	84	84
Glen Street & Town Path	SB	92	94	97
Glen Street & Town Path	EB	71	72	81
Glen Street & Town Path	WB	60	61	67
Near Glen Cove Avenue	NB	519	527	534
Near Glen Cove Avenue	SB	368	373	382
Near Glen Cove Avenue	EB	490	497	497
Near Glen Cove Avenue	WB	336	341	342
Dosoris Lane & Forest Avenue	NB	32	33	40
Dosoris Lane & Forest Avenue	SB	30	31	36
Dosoris Lane & Forest Avenue	EB	30	30	30
Dosoris Lane & Forest Avenue	WB	23	23	25
Glen Street & Pearsall Avenue	NB	51	52	52
Glen Street & Pearsall Avenue	SB	37	37	38
Glen Street & Pearsall Avenue	EB	46	47	53
Glen Street & Pearsall Avenue	WB	31	31	36
Near Forest Avenue	NB	434	441	445
Near Forest Avenue	SB	368	373	376

Intersection	Approach	Existing (vph)	No-Build (vph)	Build (vph)
Near Forest Avenue	EB	362	367	367
Near Forest Avenue	WB	333	338	338
Cedar Swamp Road & Glen Street	NB	551	559	559
Cedar Swamp Road & Glen Street	SB	507	515	515
Cedar Swamp Road & Glen Street	EB	442	448	451
Cedar Swamp Road & Glen Street	WB	362	367	370

APPENDIX B — EXISTING CONDITION CAPACITY ANALYSIS OUTPUT

Per-intersection HCM 6 solver output for the Existing No-Build condition (current-year volumes; volumes grown to opening year at the §3.1 background-growth rate for No-Build comparison).

Intersection	Approach	Vol (vph)	v/c	Delay (s)	LOS
Glen Cove Avenue & Charles Street	NB	499	0.62	22.3	C
Glen Cove Avenue & Charles Street	SB	486	0.60	21.9	C
Glen Cove Avenue & Charles Street	EB	373	0.46	19.1	B
Glen Cove Avenue & Charles Street	WB	381	0.47	19.2	B
Glen Cove Avenue & Bridge Street	NB	90	0.11	14.6	B
Glen Cove Avenue & Bridge Street	SB	79	0.10	14.5	B
Glen Cove Avenue & Bridge Street	EB	82	0.10	14.5	B
Glen Cove Avenue & Bridge Street	WB	58	0.07	14.2	B
Brewster Street & Herb Hill Road	NB	493	0.61	22.1	C
Brewster Street & Herb Hill Road	SB	469	0.58	21.4	C
Brewster Street & Herb Hill Road	EB	423	0.52	20.2	C
Brewster Street & Herb Hill Road	WB	354	0.44	18.7	B
Glen Cove Avenue & Charles Street	NB	537	0.66	23.7	C
Glen Cove Avenue & Charles Street	SB	448	0.55	20.8	C
Glen Cove Avenue & Charles Street	EB	419	0.52	20.1	C
Glen Cove Avenue & Charles Street	WB	334	0.41	18.3	B
Near Charles Street	NB	39	0.05	14.0	B
Near Charles Street	SB	29	0.04	13.9	B
Near Charles Street	EB	27	0.03	13.9	B
Near Charles Street	WB	23	0.03	13.9	B
Bridge Street & School Street	NB	82	0.10	14.5	B
Bridge Street & School Street	SB	80	0.10	14.5	B
Bridge Street & School Street	EB	82	0.10	14.5	B
Bridge Street & School Street	WB	67	0.08	14.3	B
Pratt Boulevard & Glen Street	NB	308	0.38	17.8	B
Pratt Boulevard & Glen Street	SB	232	0.29	16.5	B
Pratt Boulevard & Glen Street	EB	261	0.32	17.0	B
Pratt Boulevard & Glen Street	WB	235	0.29	16.6	B
Brewster Street & School Street	NB	341	0.42	18.4	B
Brewster Street & School Street	SB	284	0.35	17.3	B
Brewster Street & School Street	EB	232	0.29	16.5	B
Brewster Street & School Street	WB	205	0.25	16.1	B
School Street & Highland Road	NB	29	0.04	13.9	B
School Street & Highland Road	SB	25	0.03	13.9	B
School Street & Highland Road	EB	24	0.03	13.9	B
School Street & Highland Road	WB	24	0.03	13.9	B

Intersection	Approach	Vol (vph)	v/c	Delay (s)	LOS
Shore Road & Glen Cove Avenue	NB	563	0.69	24.7	C
Shore Road & Glen Cove Avenue	SB	395	0.49	19.5	B
Shore Road & Glen Cove Avenue	EB	395	0.49	19.5	B
Shore Road & Glen Cove Avenue	WB	386	0.48	19.3	B
Glen Street & Pratt Boulevard	NB	93	0.12	14.6	B
Glen Street & Pratt Boulevard	SB	76	0.09	14.4	B
Glen Street & Pratt Boulevard	EB	74	0.09	14.4	B
Glen Street & Pratt Boulevard	WB	67	0.08	14.3	B
Pratt Boulevard & Town Path	NB	344	0.42	18.5	B
Pratt Boulevard & Town Path	SB	295	0.36	17.6	B
Pratt Boulevard & Town Path	EB	328	0.40	18.1	B
Pratt Boulevard & Town Path	WB	226	0.28	16.4	B
School Street & Brewster Street	NB	116	0.14	14.9	B
School Street & Brewster Street	SB	87	0.11	14.6	B
School Street & Brewster Street	EB	92	0.11	14.6	B
School Street & Brewster Street	WB	78	0.10	14.5	B
Landing Road & Cottage Row	NB	102	0.13	14.7	B
Landing Road & Cottage Row	SB	101	0.12	14.7	B
Landing Road & Cottage Row	EB	88	0.11	14.6	B
Landing Road & Cottage Row	WB	72	0.09	14.4	B
Glen Street & Town Path	NB	84	0.10	14.5	B
Glen Street & Town Path	SB	94	0.12	14.6	B
Glen Street & Town Path	EB	72	0.09	14.4	B
Glen Street & Town Path	WB	61	0.08	14.3	B
Near Glen Cove Avenue	NB	527	0.65	23.3	C
Near Glen Cove Avenue	SB	373	0.46	19.1	B
Near Glen Cove Avenue	EB	497	0.61	22.3	C
Near Glen Cove Avenue	WB	341	0.42	18.4	B
Dosoris Lane & Forest Avenue	NB	33	0.04	14.0	B
Dosoris Lane & Forest Avenue	SB	31	0.04	13.9	B
Dosoris Lane & Forest Avenue	EB	30	0.04	13.9	B
Dosoris Lane & Forest Avenue	WB	23	0.03	13.9	B
Glen Street & Pearsall Avenue	NB	52	0.06	14.2	B
Glen Street & Pearsall Avenue	SB	37	0.05	14.0	B
Glen Street & Pearsall Avenue	EB	47	0.06	14.1	B
Glen Street & Pearsall Avenue	WB	31	0.04	13.9	B
Near Forest Avenue	NB	441	0.54	20.6	C
Near Forest Avenue	SB	373	0.46	19.1	B
Near Forest Avenue	EB	367	0.45	18.9	B

Intersection	Approach	Vol (vph)	v/c	Delay (s)	LOS
Near Forest Avenue	WB	338	0.42	18.3	B
Cedar Swamp Road & Glen Street	NB	559	0.69	24.5	C
Cedar Swamp Road & Glen Street	SB	515	0.64	22.9	C
Cedar Swamp Road & Glen Street	EB	448	0.55	20.8	C
Cedar Swamp Road & Glen Street	WB	367	0.45	18.9	B

APPENDIX C — PROPOSED CONDITION CAPACITY ANALYSIS OUTPUT

Per-intersection HCM 6 solver output for the Proposed Build condition (No-Build volumes plus project external trips at the assigned distribution).

Intersection	Approach	Vol (vph)	v/c	Delay (s)	LOS
Glen Cove Avenue & Charles Street	NB	598	0.74	26.4	C
Glen Cove Avenue & Charles Street	SB	577	0.71	25.3	C
Glen Cove Avenue & Charles Street	EB	375	0.46	19.1	B
Glen Cove Avenue & Charles Street	WB	381	0.47	19.2	B
Glen Cove Avenue & Bridge Street	NB	90	0.11	14.6	B
Glen Cove Avenue & Bridge Street	SB	109	0.13	14.8	B
Glen Cove Avenue & Bridge Street	EB	148	0.18	15.3	B
Glen Cove Avenue & Bridge Street	WB	89	0.11	14.6	B
Brewster Street & Herb Hill Road	NB	559	0.69	24.5	C
Brewster Street & Herb Hill Road	SB	522	0.64	23.1	C
Brewster Street & Herb Hill Road	EB	424	0.52	20.2	C
Brewster Street & Herb Hill Road	WB	361	0.45	18.8	B
Glen Cove Avenue & Charles Street	NB	582	0.72	25.5	C
Glen Cove Avenue & Charles Street	SB	502	0.62	22.4	C
Glen Cove Avenue & Charles Street	EB	419	0.52	20.1	C
Glen Cove Avenue & Charles Street	WB	340	0.42	18.4	B
Near Charles Street	NB	54	0.07	14.2	B
Near Charles Street	SB	57	0.07	14.2	B
Near Charles Street	EB	30	0.04	13.9	B
Near Charles Street	WB	72	0.09	14.4	B
Bridge Street & School Street	NB	130	0.16	15.1	B
Bridge Street & School Street	SB	113	0.14	14.9	B
Bridge Street & School Street	EB	82	0.10	14.5	B
Bridge Street & School Street	WB	80	0.10	14.5	B
Pratt Boulevard & Glen Street	NB	308	0.38	17.8	B
Pratt Boulevard & Glen Street	SB	245	0.30	16.7	B
Pratt Boulevard & Glen Street	EB	297	0.37	17.6	B
Pratt Boulevard & Glen Street	WB	254	0.31	16.9	B
Brewster Street & School Street	NB	365	0.45	18.9	B
Brewster Street & School Street	SB	302	0.37	17.7	B
Brewster Street & School Street	EB	232	0.29	16.5	B
Brewster Street & School Street	WB	208	0.26	16.2	B
School Street & Highland Road	NB	52	0.06	14.2	B
School Street & Highland Road	SB	41	0.05	14.0	B
School Street & Highland Road	EB	24	0.03	13.9	B

Intersection	Approach	Vol (vph)	v/c	Delay (s)	LOS
School Street & Highland Road	WB	28	0.03	13.9	B
Shore Road & Glen Cove Avenue	NB	577	0.71	25.3	C
Shore Road & Glen Cove Avenue	SB	413	0.51	19.9	B
Shore Road & Glen Cove Avenue	EB	395	0.49	19.5	B
Shore Road & Glen Cove Avenue	WB	387	0.48	19.4	B
Glen Street & Pratt Boulevard	NB	93	0.12	14.6	B
Glen Street & Pratt Boulevard	SB	81	0.10	14.5	B
Glen Street & Pratt Boulevard	EB	88	0.11	14.6	B
Glen Street & Pratt Boulevard	WB	75	0.09	14.4	B
Pratt Boulevard & Town Path	NB	344	0.42	18.5	B
Pratt Boulevard & Town Path	SB	299	0.37	17.6	B
Pratt Boulevard & Town Path	EB	341	0.42	18.4	B
Pratt Boulevard & Town Path	WB	235	0.29	16.6	B
School Street & Brewster Street	NB	128	0.16	15.1	B
School Street & Brewster Street	SB	97	0.12	14.7	B
School Street & Brewster Street	EB	92	0.11	14.6	B
School Street & Brewster Street	WB	80	0.10	14.5	B
Landing Road & Cottage Row	NB	114	0.14	14.9	B
Landing Road & Cottage Row	SB	111	0.14	14.9	B
Landing Road & Cottage Row	EB	88	0.11	14.6	B
Landing Road & Cottage Row	WB	73	0.09	14.4	B
Glen Street & Town Path	NB	84	0.10	14.5	B
Glen Street & Town Path	SB	97	0.12	14.7	B
Glen Street & Town Path	EB	81	0.10	14.5	B
Glen Street & Town Path	WB	67	0.08	14.3	B
Near Glen Cove Avenue	NB	534	0.66	23.5	C
Near Glen Cove Avenue	SB	382	0.47	19.2	B
Near Glen Cove Avenue	EB	497	0.61	22.3	C
Near Glen Cove Avenue	WB	342	0.42	18.4	B
Dosoris Lane & Forest Avenue	NB	40	0.05	14.0	B
Dosoris Lane & Forest Avenue	SB	36	0.04	14.0	B
Dosoris Lane & Forest Avenue	EB	30	0.04	13.9	B
Dosoris Lane & Forest Avenue	WB	25	0.03	13.9	B
Glen Street & Pearsall Avenue	NB	52	0.06	14.2	B
Glen Street & Pearsall Avenue	SB	38	0.05	14.0	B
Glen Street & Pearsall Avenue	EB	53	0.07	14.2	B
Glen Street & Pearsall Avenue	WB	36	0.04	14.0	B
Near Forest Avenue	NB	445	0.55	20.7	C
Near Forest Avenue	SB	376	0.46	19.1	B

Intersection	Approach	Vol (vph)	v/c	Delay (s)	LOS
Near Forest Avenue	EB	367	0.45	18.9	B
Near Forest Avenue	WB	338	0.42	18.3	B
Cedar Swamp Road & Glen Street	NB	559	0.69	24.5	C
Cedar Swamp Road & Glen Street	SB	515	0.64	22.9	C
Cedar Swamp Road & Glen Street	EB	451	0.56	20.9	C
Cedar Swamp Road & Glen Street	WB	370	0.46	19.0	B

APPENDIX D — CRASH ANALYSIS DIAGRAMS AND TABLES

Crash analysis diagrams and tables are not produced by this screening analysis. When the §4.0 crash-analysis section is expanded for formal submittal, the following NYSDOT forms are required: TE-156a (Collision Diagram), TE-164a (Safety Benefits Evaluation Form), TE-204a (Accident Rate Summary), and TE-213 (HAL / PIL / SDL Screening Output). SIMS output and statewide-average rate references should be obtained from the controlling Regional Traffic Office per HDM Chapter 5 §5.3.4 and the NYSDOT Office of Modal Safety statewide rate tables (<https://www.dot.ny.gov/divisions/operating/osss/highway/accident-rates>).

This report is based on the NYSDOT TIS Shell revised on 9/16/2014.

FOUR-STEP TRAVEL DEMAND MODEL

Off-site project trips are distributed and assigned with the four-step urban transportation modeling process (FHWA; NCHRP Report 716, Travel Demand Forecasting). Each step below shows what the engine computed for this site.

Step 1 — Trip Generation

Public-data screening trip rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) applied to the proposed land use give the site's productions.

Land use	LU 820 — Shopping Center / Retail
Size	85 1,000 sqft GLA
Daily trips (productions)	6,800
PM peak-hour trips	680 (326 in / 354 out)

Step 2 — Trip Distribution (Gravity Model)

A production-constrained gravity model allocates trips to surrounding signals: $T_j = P \cdot (A_j \cdot F_j) / \sum (A_x \cdot F_x)$. Attractiveness A_j is each signal's through-volume; the friction factor F_j is the NCHRP-716 gamma function $F = a \cdot t^b \cdot e^{(c \cdot t)}$ (home-based-work: $a=28507$, $b=-0.02$, $c=-0.123$) on the travel time t to the signal.

Signal (attraction zone)	Dist mi	Time min	PM trips	Share
Glen Cove Avenue & Charles Street	0.02	0.1	191	17.4%
Glen Cove Avenue & Bridge Street	0.10	0.2	126	11.5%
Brewster Street & Herb Hill Road	0.10	0.2	126	11.5%
Glen Cove Avenue & Charles Street	0.14	0.3	104	9.5%
Near Charles Street	0.16	0.4	94	8.5%
Bridge Street & School Street	0.16	0.4	94	8.5%
Pratt Boulevard & Glen Street	0.22	0.5	69	6.3%
Brewster Street & School Street	0.30	0.7	46	4.2%
School Street & Highland Road	0.31	0.7	44	4.0%
Shore Road & Glen Cove Avenue	0.37	0.9	33	3.0%
Glen Street & Pratt Boulevard	0.41	1.0	27	2.5%
Pratt Boulevard & Town Path	0.42	1.0	26	2.4%

Top 12 distribution zones by assigned PM-peak trips shown; the full set is in the capacity appendix.

Step 3 — Mode Choice

Auto-mode share 44% applied via a binary logit $P(\text{auto}) = 1 / (1 + e^{-(ASC - \lambda \cdot \Delta GC)})$ calibrated to the metro's measured auto-mode share (ACS B08301) and shifted by site urbanity (local density), so a denser, more transit-served site splits further from auto than a greenfield parcel in the same metro. The remaining 56% of trips (transit, walking, cycling) do not load the off-site roadway network; only auto trips are carried into Step 4.

Step 4 — Route Assignment

A capacity-constrained assignment using the BPR volume-delay function $t = t_0 \cdot [1 + 0.15 \cdot (v/c)^4]$ iteratively shifts trips away from over-capacity signals toward less-congested alternatives until the assignment is congestion-consistent. Highest post-build v/c among study signals: 1.07. Per-signal v/c , delay, LOS, and queue are in the capacity appendix. (Road-network corridor routing was not available for this region; per-signal capacity-constrained assignment was used.)

APPENDIX — INTERSECTION CAPACITY ANALYSIS WORKSHEETS

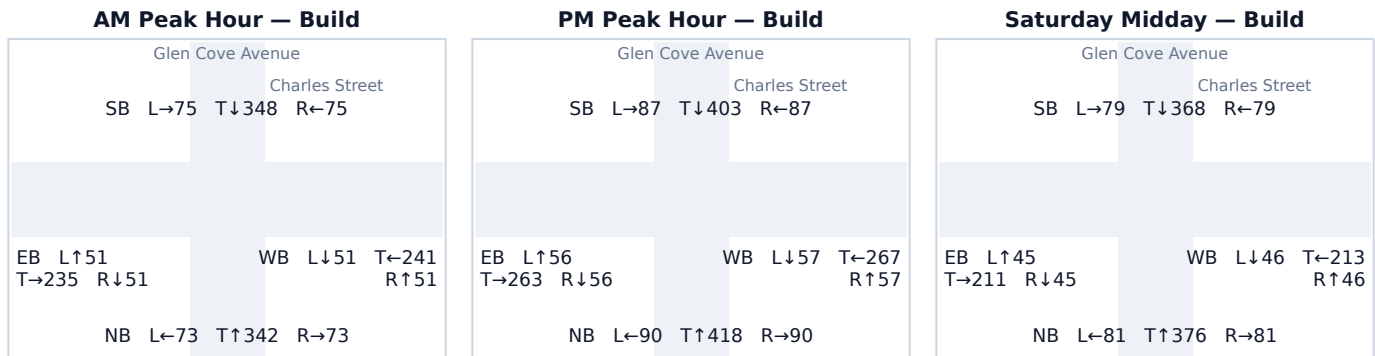
One worksheet per study intersection: a turning-movement diagram for each analyzed peak period plus the per-approach HCM capacity table. Analysis follows the Highway Capacity Manual 6th Edition signalized-intersection method; v/c, control delay (sec/veh), Level of Service, and 95th-percentile back-of-queue (ft) are reported for the Existing (No-Build) and Build conditions.

Background turning-movement volumes in the diagrams are distributed from each approach total using an estimated 15/70/15 (Left/Through/Right) split. Project-trip movements are assigned geometrically from the study's directional trip distribution (see each worksheet's Affected movements table). Replace both with measured turning-movement counts (TMCs) before a formal submittal.

A.1 Glen Cove Avenue & Charles Street

Signal ID	new-york-13590
Location	NE New York-Newark-Jersey City · 0.02 mi from site
Added PM peak trips	191
Existing / No-Build (PM)	LOS D · 48.4 s/veh · v/c 0.97
Build (PM)	LOS E · 77.4 s/veh · v/c 1.07
Δ control delay	+29.0 s/veh (LOS change)
Mitigation	Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	499	100	598	0.62 → 0.74	22.3 → 26.4	C → C	508
SB	486	90	577	0.60 → 0.71	21.9 → 25.3	C → C	481
EB	373	1	375	0.46 → 0.46	19.1 → 19.1	B → B	268
WB	381	0	381	0.47 → 0.47	19.2 → 19.2	B → B	274

Affected movements (PM peak project trips)

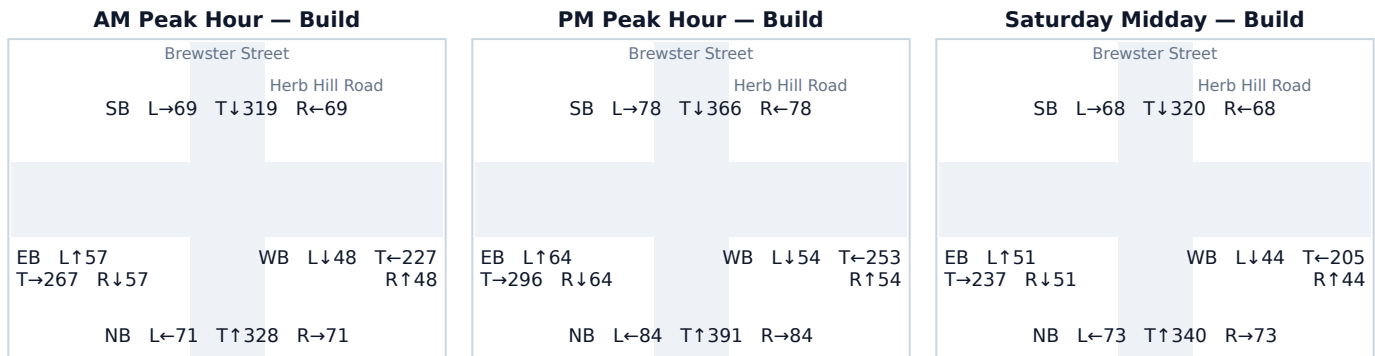
Movement	Project trips	% of project trips
NB Through	98	51%
SB Through	90	47%
NB Left	2	1%
EB Right	1	1%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.2 Brewster Street & Herb Hill Road

Signal ID	new-york-12426
Location	NE New York-Newark-Jersey City · 0.10 mi from site
Added PM peak trips	126
Existing / No-Build (PM)	LOS D · 48.4 s/veh · v/c 0.97
Build (PM)	LOS E · 65.8 s/veh · v/c 1.04
Δ control delay	+17.4 s/veh (LOS change)
Mitigation	Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	493	66	559	0.61 → 0.69	22.1 → 24.5	C → C	459
SB	469	53	522	0.58 → 0.64	21.4 → 23.1	C → C	417
EB	423	0	424	0.52 → 0.52	20.2 → 20.2	C → C	314
WB	354	7	361	0.44 → 0.45	18.7 → 18.8	B → B	256

Affected movements (PM peak project trips)

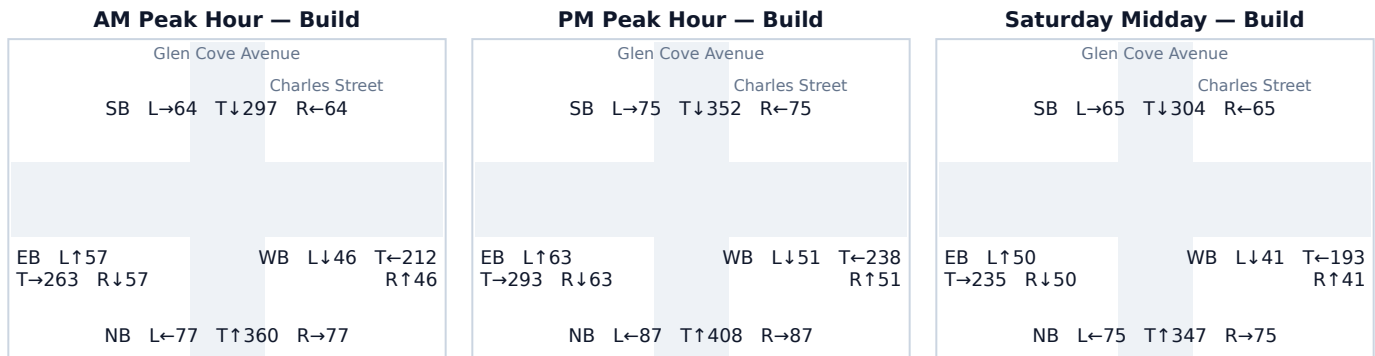
Movement	Project trips	% of project trips
NB Through	58	46%
SB Through	53	42%
NB Right	8	6%
WB Left	7	6%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.3 Glen Cove Avenue & Charles Street

Signal ID	new-york-13514
Location	NE New York-Newark-Jersey City · 0.14 mi from site
Added PM peak trips	104
Existing / No-Build (PM)	LOS D · 48.4 s/veh · v/c 0.97
Build (PM)	LOS E · 62.3 s/veh · v/c 1.02
Δ control delay	+13.9 s/veh (LOS change)
Mitigation	Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	537	45	582	0.66 → 0.72	23.7 → 25.5	C → C	488
SB	448	54	502	0.55 → 0.62	20.8 → 22.4	C → C	395
EB	419	0	419	0.52 → 0.52	20.1 → 20.1	C → C	310
WB	334	5	340	0.41 → 0.42	18.3 → 18.4	B → B	237

Affected movements (PM peak project trips)

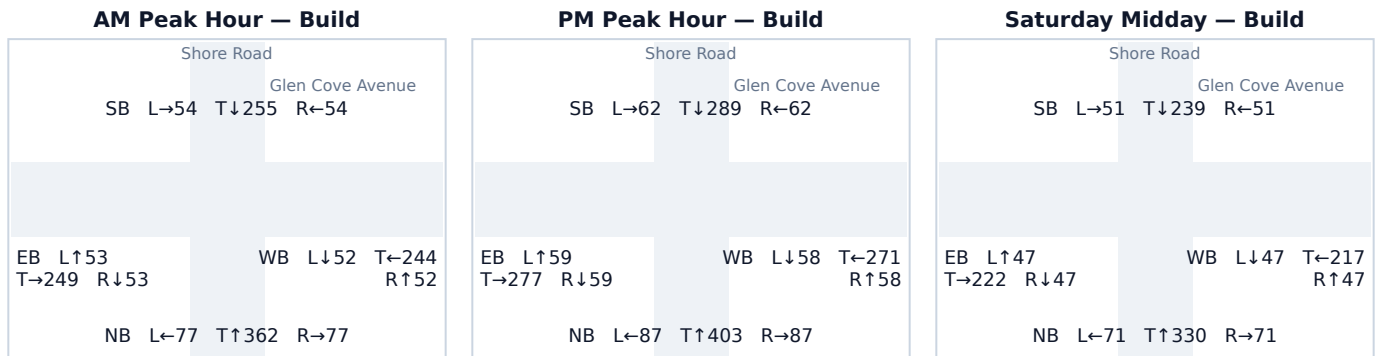
Movement	Project trips	% of project trips
SB Through	48	46%
NB Through	45	43%
SB Left	6	6%
WB Right	5	5%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.4 Shore Road & Glen Cove Avenue

Signal ID	new-york-13099
Location	NE New York-Newark-Jersey City · 0.37 mi from site
Added PM peak trips	33
Existing / No-Build (PM)	LOS D · 48.4 s/veh · v/c 0.97
Build (PM)	LOS D · 52.5 s/veh · v/c 0.98
Δ control delay	+4.1 s/veh
Mitigation	Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	563	14	577	0.69 → 0.71	24.7 → 25.3	C → C	482
SB	395	17	413	0.49 → 0.51	19.5 → 19.9	B → B	304
EB	395	0	395	0.49 → 0.49	19.5 → 19.5	B → B	287
WB	386	2	387	0.48 → 0.48	19.3 → 19.4	B → B	280

Affected movements (PM peak project trips)

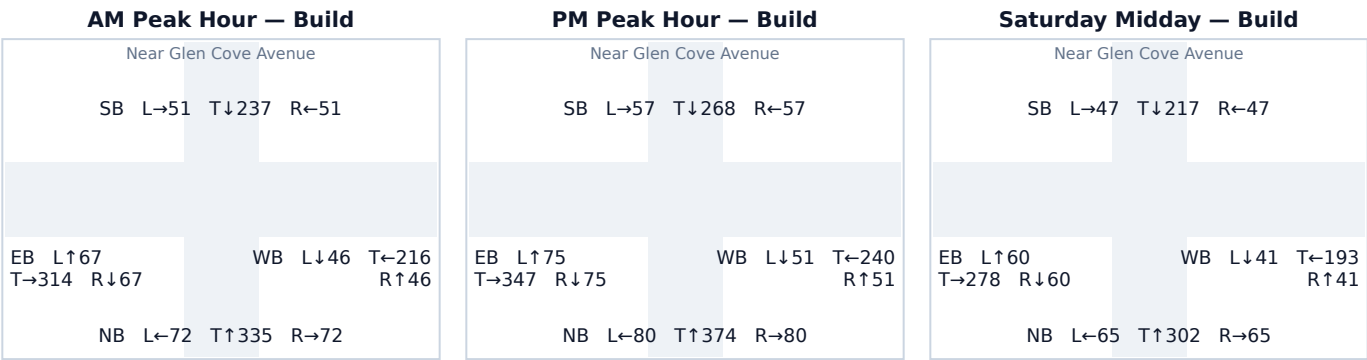
Movement	Project trips	% of project trips
SB Through	15	45%
NB Through	14	42%
SB Left	2	6%
WB Right	2	6%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.5 Near Glen Cove Avenue

Signal ID	new-york-13589
Location	NE New York-Newark-Jersey City · 0.50 mi from site
Added PM peak trips	17
Existing / No-Build (PM)	LOS D · 48.4 s/veh · v/c 0.97
Build (PM)	LOS D · 50.4 s/veh · v/c 0.98
Δ control delay	+2.0 s/veh
Mitigation	Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	527	7	534	0.65 → 0.66	23.3 → 23.5	C → C	430
SB	373	9	382	0.46 → 0.47	19.1 → 19.2	B → B	275
EB	497	0	497	0.61 → 0.61	22.3 → 22.3	C → C	390
WB	341	1	342	0.42 → 0.42	18.4 → 18.4	B → B	240

Affected movements (PM peak project trips)

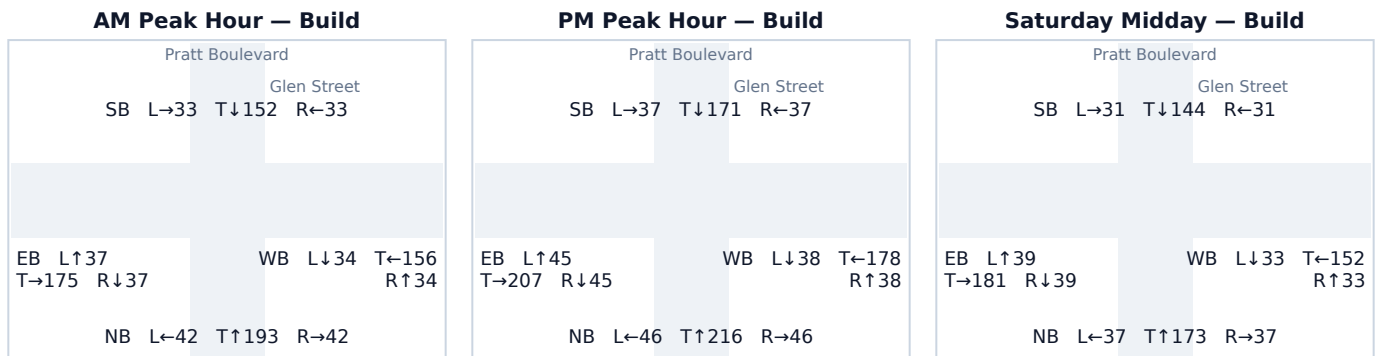
Movement	Project trips	% of project trips
SB Through	8	47%
NB Through	7	41%
SB Left	1	6%
WB Right	1	6%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.6 Pratt Boulevard & Glen Street

Signal ID	new-york-18387
Location	NE New York-Newark-Jersey City · 0.22 mi from site
Added PM peak trips	69
Existing / No-Build (PM)	LOS C · 21.3 s/veh · v/c 0.58
Build (PM)	LOS C · 22.3 s/veh · v/c 0.61
Δ control delay	+1.0 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	308	0	308	0.38 → 0.38	17.8 → 17.8	B → B	211
SB	232	13	245	0.29 → 0.30	16.5 → 16.7	B → B	161
EB	261	36	297	0.32 → 0.37	17.0 → 17.6	B → B	202
WB	235	20	254	0.29 → 0.31	16.6 → 16.9	B → B	168

Affected movements (PM peak project trips)

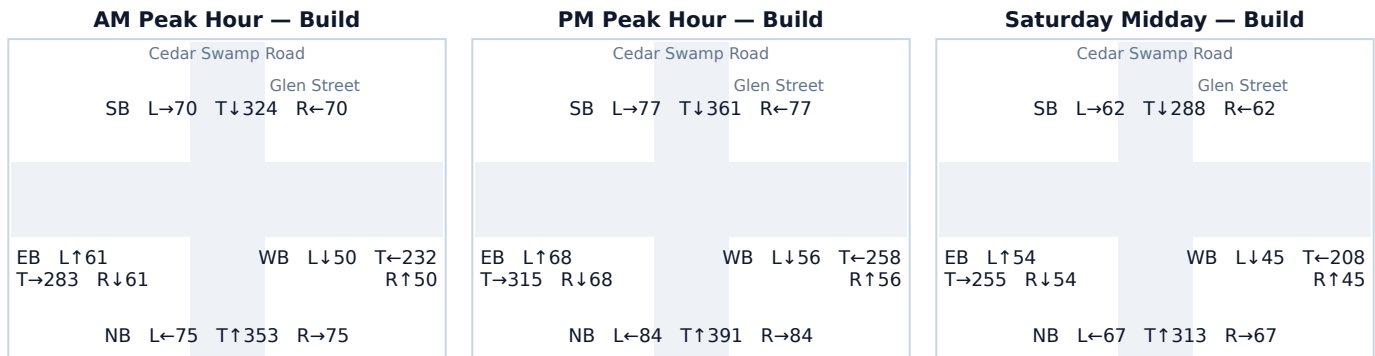
Movement	Project trips	% of project trips
EB Through	21	30%
WB Through	20	29%
EB Left	15	22%
SB Right	13	19%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.7 Cedar Swamp Road & Glen Street

Signal ID	new-york-13215
Location	NE New York-Newark-Jersey City · 0.72 mi from site
Added PM peak trips	6
Existing / No-Build (PM)	LOS E · 69.9 s/veh · v/c 1.05
Build (PM)	LOS E · 70.9 s/veh · v/c 1.05
Δ control delay	+1.0 s/veh
Mitigation	Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	559	0	559	0.69 → 0.69	24.5 → 24.5	C → C	460
SB	515	0	515	0.64 → 0.64	22.9 → 22.9	C → C	409
EB	448	3	451	0.55 → 0.56	20.8 → 20.9	C → C	341
WB	367	3	370	0.45 → 0.46	18.9 → 19.0	B → B	264

Affected movements (PM peak project trips)

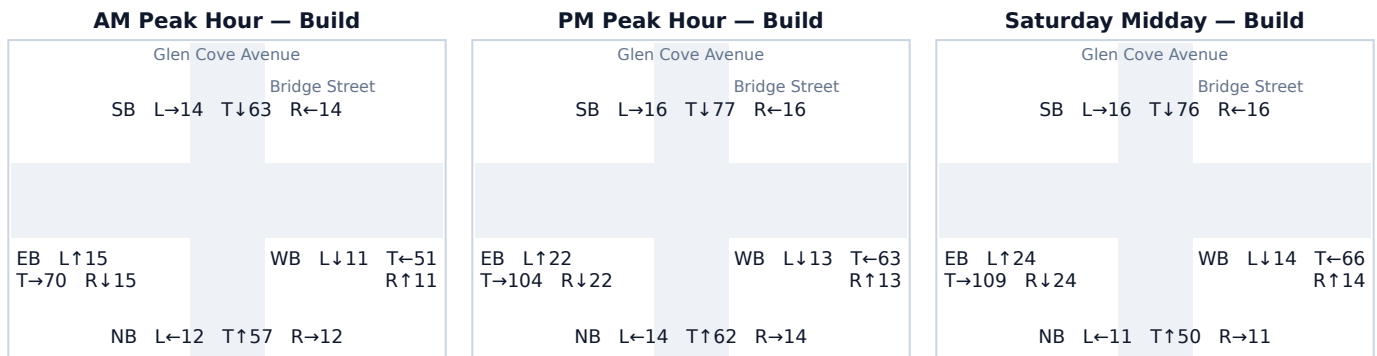
Movement	Project trips	% of project trips
EB Through	3	50%
WB Through	3	50%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.8 Glen Cove Avenue & Bridge Street

Signal ID	new-york-12469
Location	NE New York-Newark-Jersey City · 0.10 mi from site
Added PM peak trips	126
Existing / No-Build (PM)	LOS B · 15.2 s/veh · v/c 0.17
Build (PM)	LOS B · 16.0 s/veh · v/c 0.24
Δ control delay	+0.8 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	90	0	90	0.11 → 0.11	14.6 → 14.6	B → B	54
SB	79	29	109	0.10 → 0.13	14.5 → 14.8	B → B	66
EB	82	66	148	0.10 → 0.18	14.5 → 15.3	B → B	91
WB	58	31	89	0.07 → 0.11	14.2 → 14.6	B → B	53

Affected movements (PM peak project trips)

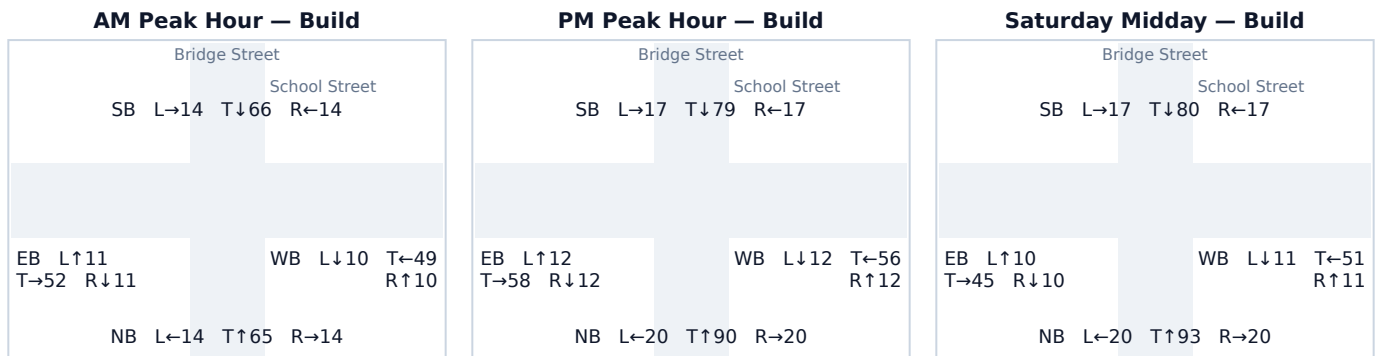
Movement	Project trips	% of project trips
EB Through	34	27%
EB Left	32	25%
WB Through	31	25%
SB Right	29	23%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.9 Bridge Street & School Street

Signal ID	new-york-13429
Location	NE New York-Newark-Jersey City · 0.16 mi from site 94
Added PM peak trips	LOS B · 15.2 s/veh · v/c 0.17
Existing / No-Build (PM)	LOS B · 15.8 s/veh · v/c 0.22
Build (PM)	+0.6 s/veh
Δ control delay	No mitigation required — projected delay change is below the City's 5-second TIS threshold.
Mitigation	

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	82	49	130	0.10 → 0.16	14.5 → 15.1	B → B	80
SB	80	33	113	0.10 → 0.14	14.5 → 14.9	B → B	68
EB	82	0	82	0.10 → 0.10	14.5 → 14.5	B → B	49
WB	67	12	80	0.08 → 0.10	14.3 → 14.5	B → B	47

Affected movements (PM peak project trips)

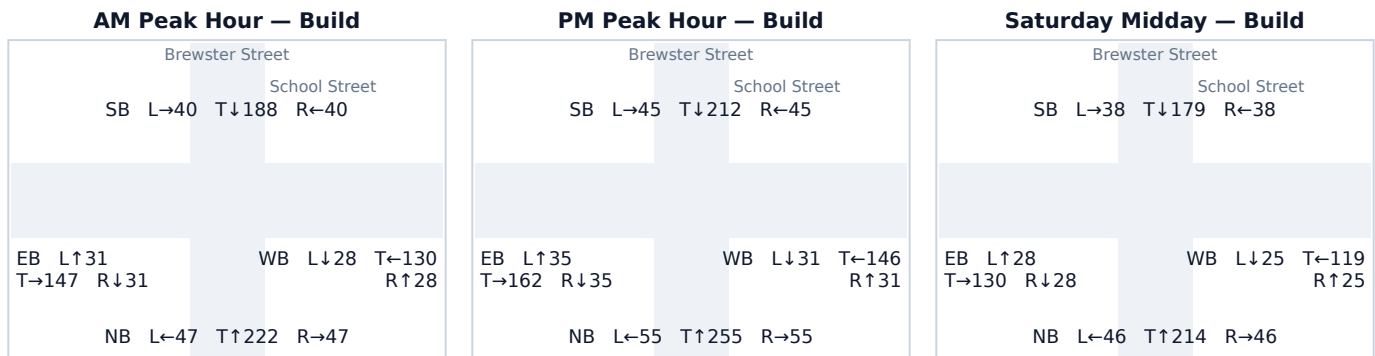
Movement	Project trips	% of project trips
NB Through	35	37%
SB Through	33	35%
NB Right	14	15%
WB Left	12	13%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.10 Brewster Street & School Street

Signal ID	new-york-18023
Location	NE New York-Newark-Jersey City · 0.30 mi from site
Added PM peak trips	46
Existing / No-Build (PM)	LOS C · 21.7 s/veh · v/c 0.59
Build (PM)	LOS C · 22.3 s/veh · v/c 0.61
Δ control delay	+0.6 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	341	24	365	0.42 → 0.45	18.4 → 18.9	B → B	259
SB	284	19	302	0.35 → 0.37	17.3 → 17.7	B → B	206
EB	232	0	232	0.29 → 0.29	16.5 → 16.5	B → B	151
WB	205	3	208	0.25 → 0.26	16.1 → 16.2	B → B	133

Affected movements (PM peak project trips)

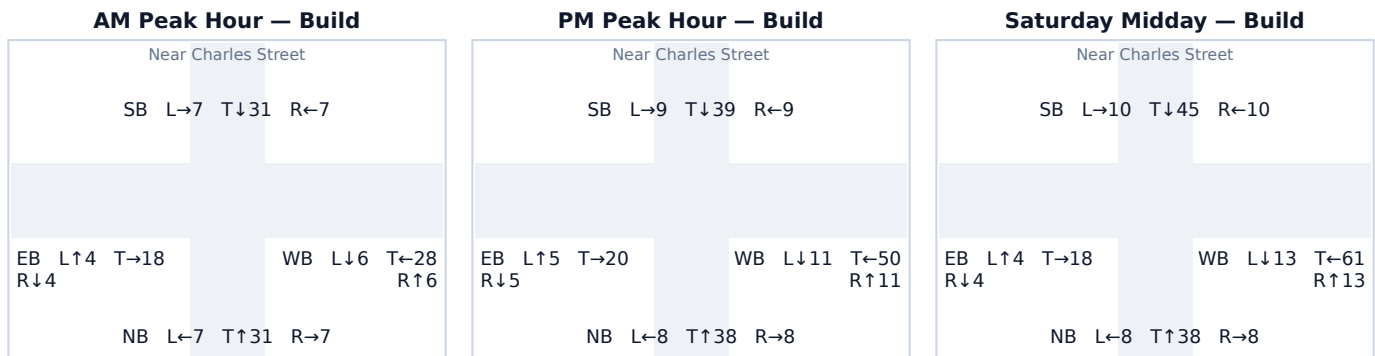
Movement	Project trips	% of project trips
NB Through	20	43%
SB Through	19	41%
NB Right	4	9%
WB Left	3	7%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.11 Near Charles Street

Signal ID	new-york-12874
Location	NE New York-Newark-Jersey City · 0.16 mi from site
Added PM peak trips	94
Existing / No-Build (PM)	LOS B · 14.2 s/veh · v/c 0.07
Build (PM)	LOS B · 14.7 s/veh · v/c 0.12
Δ control delay	+0.5 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	39	15	54	0.05 → 0.07	14.0 → 14.2	B → B	32
SB	29	28	57	0.04 → 0.07	13.9 → 14.2	B → B	33
EB	27	2	30	0.03 → 0.04	13.9 → 13.9	B → B	17
WB	23	49	72	0.03 → 0.09	13.9 → 14.4	B → B	43

Affected movements (PM peak project trips)

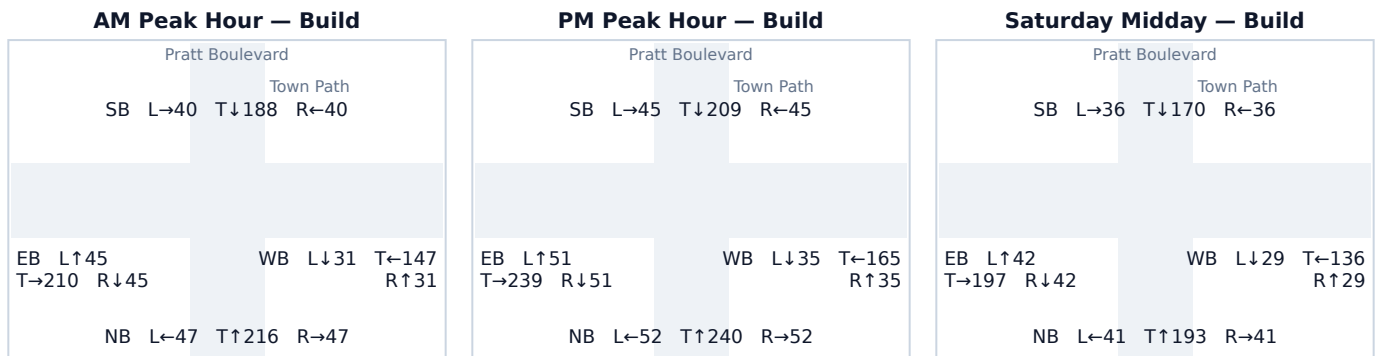
Movement	Project trips	% of project trips
WB Right	30	32%
SB Left	28	30%
WB Left	16	17%
NB Right	15	16%
WB Through	3	3%
EB Through	2	2%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.12 Pratt Boulevard & Town Path

Signal ID	new-york-13270
Location	NE New York-Newark-Jersey City · 0.42 mi from site
Added PM peak trips	26
Existing / No-Build (PM)	LOS C · 23.6 s/veh · v/c 0.66
Build (PM)	LOS C · 24.1 s/veh · v/c 0.68
Δ control delay	+0.5 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	344	0	344	0.42 → 0.42	18.5 → 18.5	B → B	241
SB	295	3	299	0.36 → 0.37	17.6 → 17.6	B → B	203
EB	328	14	341	0.40 → 0.42	18.1 → 18.4	B → B	239
WB	226	9	235	0.28 → 0.29	16.4 → 16.6	B → B	153

Affected movements (PM peak project trips)

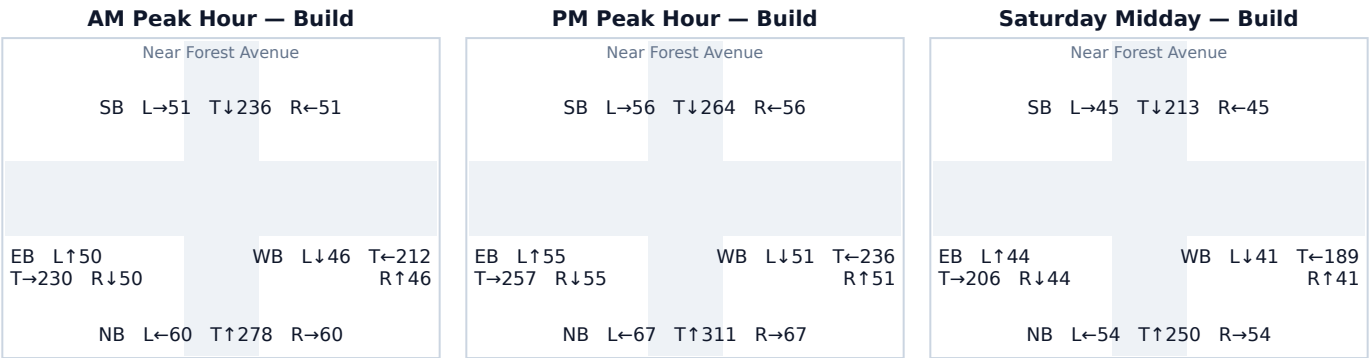
Movement	Project trips	% of project trips
EB Through	10	38%
WB Through	9	35%
EB Left	4	15%
SB Right	3	12%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.13 Near Forest Avenue

Signal ID	new-york-14799
Location	NE New York-Newark-Jersey City · 0.67 mi from site
Added PM peak trips	7
Existing / No-Build (PM)	LOS C · 32.4 s/veh · v/c 0.84
Build (PM)	LOS C · 32.7 s/veh · v/c 0.85
Δ control delay	+0.3 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	441	4	445	0.54 → 0.55	20.6 → 20.7	C → C	335
SB	373	2	376	0.46 → 0.46	19.1 → 19.1	B → B	270
EB	367	0	367	0.45 → 0.45	18.9 → 18.9	B → B	262
WB	338	1	338	0.42 → 0.42	18.3 → 18.3	B → B	236

Affected movements (PM peak project trips)

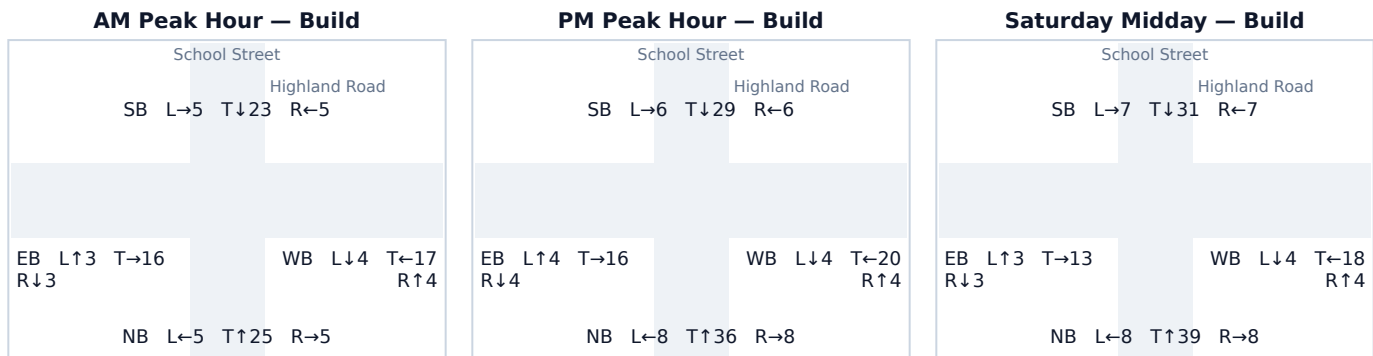
Movement	Project trips	% of project trips
NB Through	3	43%
SB Through	2	29%
NB Right	1	14%
WB Left	1	14%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.14 School Street & Highland Road

Signal ID	new-york-12696
Location	NE New York-Newark-Jersey City · 0.31 mi from site
Added PM peak trips	44
Existing / No-Build (PM)	LOS B · 14.1 s/veh · v/c 0.06
Build (PM)	LOS B · 14.3 s/veh · v/c 0.08
Δ control delay	+0.2 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	29	23	52	0.04 → 0.06	13.9 → 14.2	B → B	30
SB	25	16	41	0.03 → 0.05	13.9 → 14.0	B → B	24
EB	24	0	24	0.03 → 0.03	13.9 → 13.9	B → B	14
WB	24	5	28	0.03 → 0.03	13.9 → 13.9	B → B	16

Affected movements (PM peak project trips)

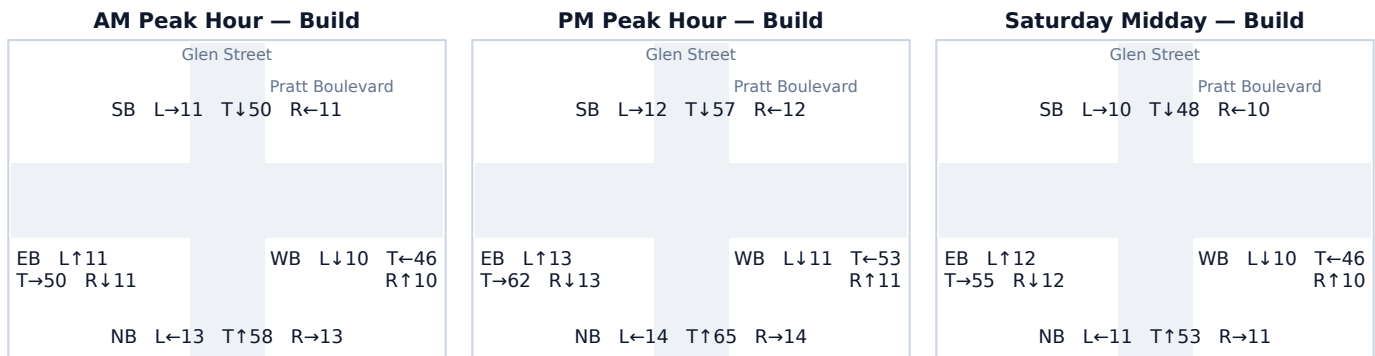
Movement	Project trips	% of project trips
NB Through	18	41%
SB Through	16	36%
NB Right	5	11%
WB Left	5	11%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.15 Glen Street & Pratt Boulevard

Signal ID	new-york-18321
Location	NE New York-Newark-Jersey City · 0.41 mi from site
Added PM peak trips	27
Existing / No-Build (PM)	LOS B · 15.2 s/veh · v/c 0.17
Build (PM)	LOS B · 15.4 s/veh · v/c 0.19
Δ control delay	+0.2 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	93	0	93	0.12 → 0.12	14.6 → 14.6	B → B	56
SB	76	5	81	0.09 → 0.10	14.4 → 14.5	B → B	48
EB	74	14	88	0.09 → 0.11	14.4 → 14.6	B → B	52
WB	67	8	75	0.08 → 0.09	14.3 → 14.4	B → B	45

Affected movements (PM peak project trips)

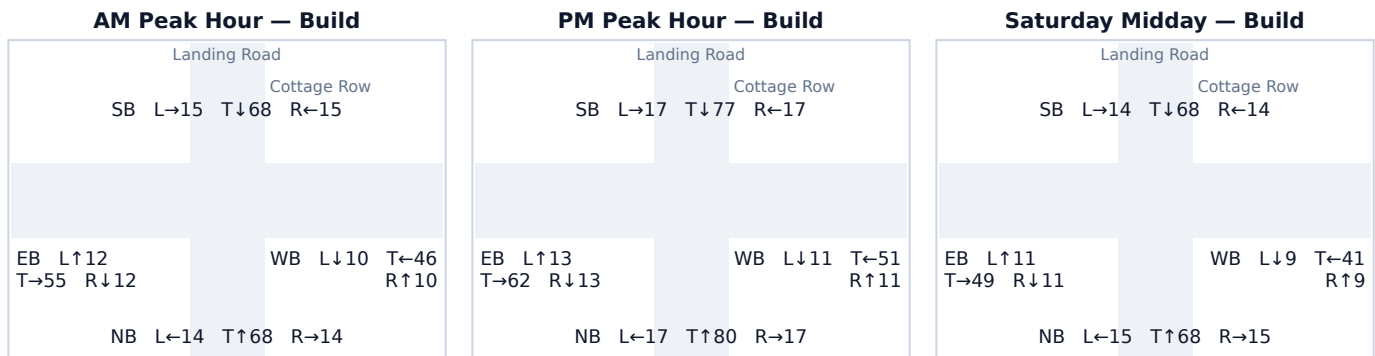
Movement	Project trips	% of project trips
EB Through	9	33%
WB Through	8	30%
EB Left	5	19%
SB Right	5	19%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.16 Landing Road & Cottage Row

Signal ID	new-york-16650
Location	NE New York-Newark-Jersey City · 0.44 mi from site
Added PM peak trips	23
Existing / No-Build (PM)	LOS B · 15.5 s/veh · v/c 0.20
Build (PM)	LOS B · 15.7 s/veh · v/c 0.21
Δ control delay	+0.2 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	102	12	114	0.13 → 0.14	14.7 → 14.9	B → B	69
SB	101	10	111	0.12 → 0.14	14.7 → 14.9	B → B	67
EB	88	0	88	0.11 → 0.11	14.6 → 14.6	B → B	53
WB	72	1	73	0.09 → 0.09	14.4 → 14.4	B → B	43

Affected movements (PM peak project trips)

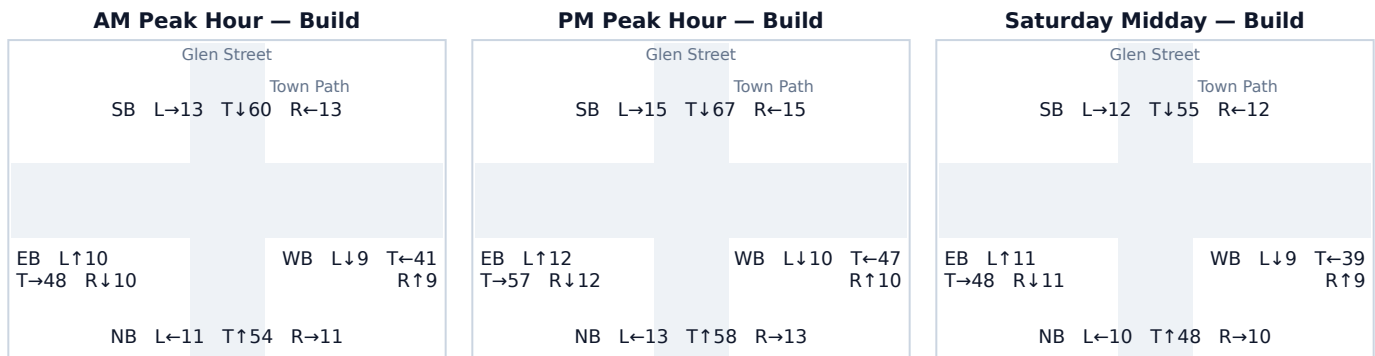
Movement	Project trips	% of project trips
NB Through	11	48%
SB Through	10	43%
NB Right	1	4%
WB Left	1	4%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.17 Glen Street & Town Path

Signal ID	new-york-13271
Location	NE New York-Newark-Jersey City · 0.50 mi from site
Added PM peak trips	17
Existing / No-Build (PM)	LOS B · 15.2 s/veh · v/c 0.17
Build (PM)	LOS B · 15.3 s/veh · v/c 0.18
Δ control delay	+0.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	84	0	84	0.10 → 0.10	14.5 → 14.5	B → B	50
SB	94	3	97	0.12 → 0.12	14.6 → 14.7	B → B	58
EB	72	9	81	0.09 → 0.10	14.4 → 14.5	B → B	48
WB	61	5	67	0.08 → 0.08	14.3 → 14.3	B → B	39

Affected movements (PM peak project trips)

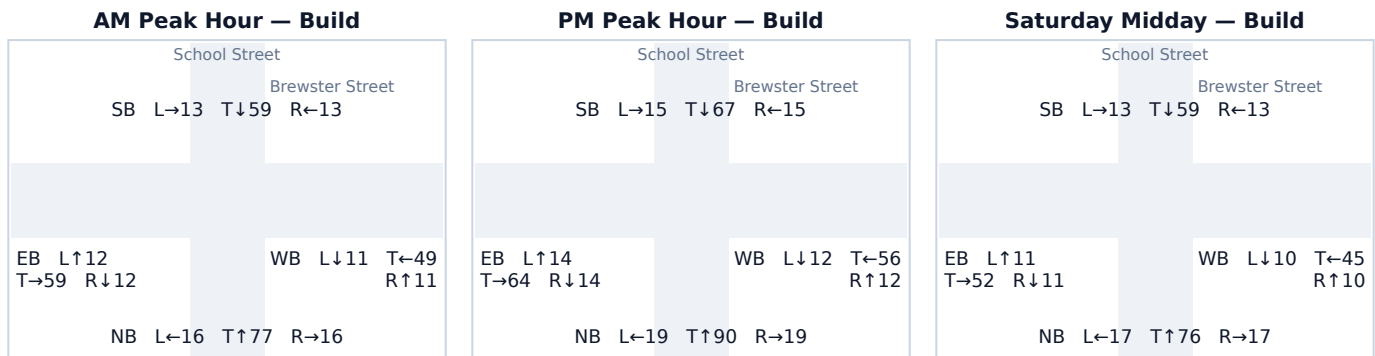
Movement	Project trips	% of project trips
EB Through	6	35%
WB Through	5	29%
EB Left	3	18%
SB Right	3	18%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.18 School Street & Brewster Street

Signal ID	new-york-13699
Location	NE New York-Newark-Jersey City · 0.43 mi from site
Added PM peak trips	25
Existing / No-Build (PM)	LOS B · 15.6 s/veh · v/c 0.21
Build (PM)	LOS B · 15.7 s/veh · v/c 0.22
Δ control delay	+0.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	116	13	128	0.14 → 0.16	14.9 → 15.1	B → B	78
SB	87	10	97	0.11 → 0.12	14.6 → 14.7	B → B	58
EB	92	0	92	0.11 → 0.11	14.6 → 14.6	B → B	55
WB	78	2	80	0.10 → 0.10	14.5 → 14.5	B → B	47

Affected movements (PM peak project trips)

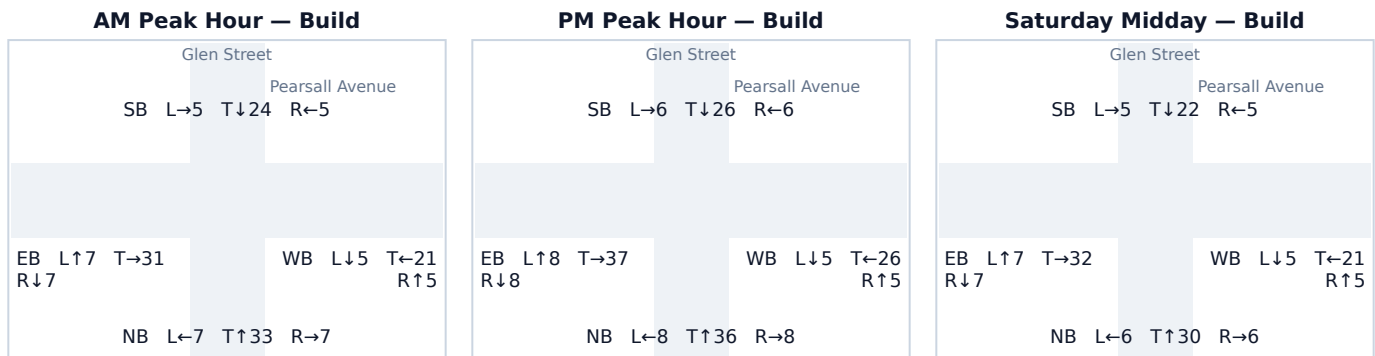
Movement	Project trips	% of project trips
NB Through	11	44%
SB Through	10	40%
NB Right	2	8%
WB Left	2	8%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.19 Glen Street & Pearsall Avenue

Signal ID	new-york-13501
Location	NE New York-Newark-Jersey City · 0.58 mi from site 11
Added PM peak trips	LOS B · 14.4 s/veh · v/c 0.09
Existing / No-Build (PM)	LOS B · 14.5 s/veh · v/c 0.10
Build (PM)	+0.1 s/veh
Δ control delay	No mitigation required — projected delay change is below the City's 5-second TIS threshold.
Mitigation	

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	52	0	52	0.06 → 0.06	14.2 → 14.2	B → B	31
SB	37	1	38	0.05 → 0.05	14.0 → 14.0	B → B	22
EB	47	6	53	0.06 → 0.07	14.1 → 14.2	B → B	31
WB	31	4	36	0.04 → 0.04	13.9 → 14.0	B → B	21

Affected movements (PM peak project trips)

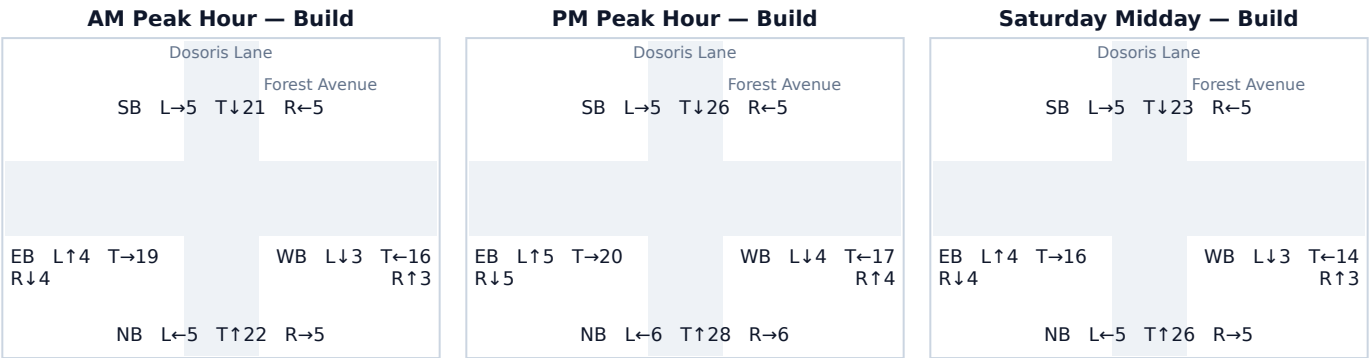
Movement	Project trips	% of project trips
EB Through	5	45%
WB Through	4	36%
EB Left	1	9%
SB Right	1	9%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.20 Dosoris Lane & Forest Avenue

Signal ID	new-york-14798
Location	NE New York-Newark-Jersey City · 0.54 mi from site
Added PM peak trips	14
Existing / No-Build (PM)	LOS B · 14.2 s/veh · v/c 0.06
Build (PM)	LOS B · 14.2 s/veh · v/c 0.07
Δ control delay	+0.0 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	33	7	40	0.04 → 0.05	14.0 → 14.0	B → B	23
SB	31	6	36	0.04 → 0.04	13.9 → 14.0	B → B	21
EB	30	0	30	0.04 → 0.04	13.9 → 13.9	B → B	18
WB	23	1	25	0.03 → 0.03	13.9 → 13.9	B → B	14

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	6	43%
SB Through	6	43%
NB Right	1	7%
WB Left	1	7%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.